

Anshika Shukla

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Education

Banasthali Vidyapith

B.Tech, Electronics and Communication Engineering (ECE)

2023 – 2027

CGPA: 7.38/10

Experience

Technical Workshop Lead / Core Member – SAMARTHYA Electronics Club

Sept 2024 – Present

Banasthali Vidyapith, Jaipur, Rajasthan

- Designed and led 5+ IoT / embedded systems and Python coding workshops for 200+ students, boosting project completion by 40% and lab participation by 35%
- Contributed to system design and technical specification for 5+ embedded workshop modules, supporting release planning and iterative delivery under tight deadlines

Technical Skills

Domain Knowledge: Control Systems, Digital Signal Processing, Embedded Systems, Computer Networks, Transfer Function Modeling

Programming Languages: Python, C, C++, JavaScript

Computer Vision: OpenCV, MediaPipe, TensorFlow Lite, Camera Interfacing, Real-Time Inference on Embedded Hardware

Embedded / Edge Deployment: Raspberry Pi, Arduino, Model Compression, Quantization, Pruning

Engineering & Automation: Basic AutoCAD, MATLAB, Simulink, Simulator-based Testing, Basic Assembly, Wiring & Installation

CS Fundamentals: Data Structures & Algorithms, OOP

AI/ML: TensorFlow, PyTorch, XGBoost, Scikit-learn, Feature Engineering

Backend & Tools: FastAPI, REST APIs, Git, GitHub, VS Code, Linux, Jupyter, Google Colab, Overleaf

Soft Skills: Team Management, Strategic Thinking, Communication Skills, Time Management, Cross-functional Coordination, Leadership, Adaptability, Problem Solving

Projects

PitWall F1 - Interactive Race Intelligence

Jun 2026

XGBoost | FastAPI | Next.js | Three.js | FastF1 | Python

- Built a machine learning pipeline on 2,449 rows of FastF1 telemetry data; trained an XGBoost regression model reaching a mean absolute error (MAE) of 3.61 seconds for race-outcome prediction
- Designed an interactive Three.js frontend to visualize live telemetry and model predictions in a race-engineer-style dashboard

SMART-GUARD - IoT-Enabled Smart Helmet Safety System

April 2026

Arduino Uno/Nano | MPU6050 | MQ3 | IR | GPS | GSM | Flask | JavaScript

- Designed a multi-sensor embedded system on Arduino integrating MPU6050 (crash detection), MQ3 (alcohol detection), IR (helmet-wear detection), and GPS/GSM modules for real-time rider safety monitoring
- Implemented sensor data acquisition and hardware integration in Arduino IDE, with a Flask backend and JavaScript/HTML/CSS dashboard for live location tracking and automated emergency alerting
- Design patent pending; hardware and system architecture developed independently, code confidential

SatLink – AI-Powered Disaster Response Platform

Jan 2026

Python | Flask | Twilio API | Satellite Communication Module | AI Event Detection

- Built an AI-powered disaster response platform that detects fire and flood events and triggers automated SOS alerts via a satellite communication module and Twilio SMS/call integration, with real-time location tracking
- Designed a full-stack system (Flask backend, web dashboard frontend) for rapid emergency response – Top 10 nationally at HACK-CELESTIA

SignEase - ASL Recognition on Raspberry Pi

Sep 2025

TensorFlow Lite | Raspberry Pi | OpenCV | MediaPipe | CNN

- Compressed a CNN model by 81% for real-time American Sign Language recognition on embedded hardware, sustaining 24 FPS at 91% accuracy with sub-100ms inference on Raspberry Pi
- Implemented live camera interfacing and pose/hand landmark estimation with OpenCV and MediaPipe for real-time embedded computer vision
- Presented the working prototype to the Department of Telecommunications at the Sanchar Saathi Initiative event

Achievements and Hackathons

- **HACKVERSE** - 2nd Runner-Up, national-level hackathon (100+ teams) – AI content moderation system, SAFEGAZE [GitHub]
- **HACKCELESTIA** - Top 10, national-level hackathon – SatLink, AI-powered disaster response platform
- **Smart India Hackathon (SIH) 2025** - College-level shortlist for an AI-powered mission control dashboard unifying astronaut health monitoring and satellite collision prevention
- **Sanchar Saathi Initiative** Selected – Govt. of India, Dept. of Telecommunications
- **GSSoC 2026** - Open Source Contributor
- **WACIS 2025** - Poster Presentation
- **SSRN Preprint** (ID 6763558) – Brain-Computer Interface research, Preprint

Certifications and Training

- CeNSE Summer School on Semiconductor Technology – Centre for Nano Science and Engineering (CeNSE)
- AI Fundamentals: Foundations for Understanding AI - IBM
- Building Data and AI Solutions on Google Cloud - 365 Data Science (100%, Mar 2026)
- Generative AI - Microsoft and LinkedIn
- Drone Technology / Drone Training Certificate – Banasthali Vidyapith
- NASA ARSET - Remote Sensing
- IEEE Student Member (Jun 2026 - Present)